清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

أرامكو السعودية
saudi aramco

Assessment of Iron and Steel Technologies from A Global Perspective

Tsinghua University & Saudi Aramco

aramco

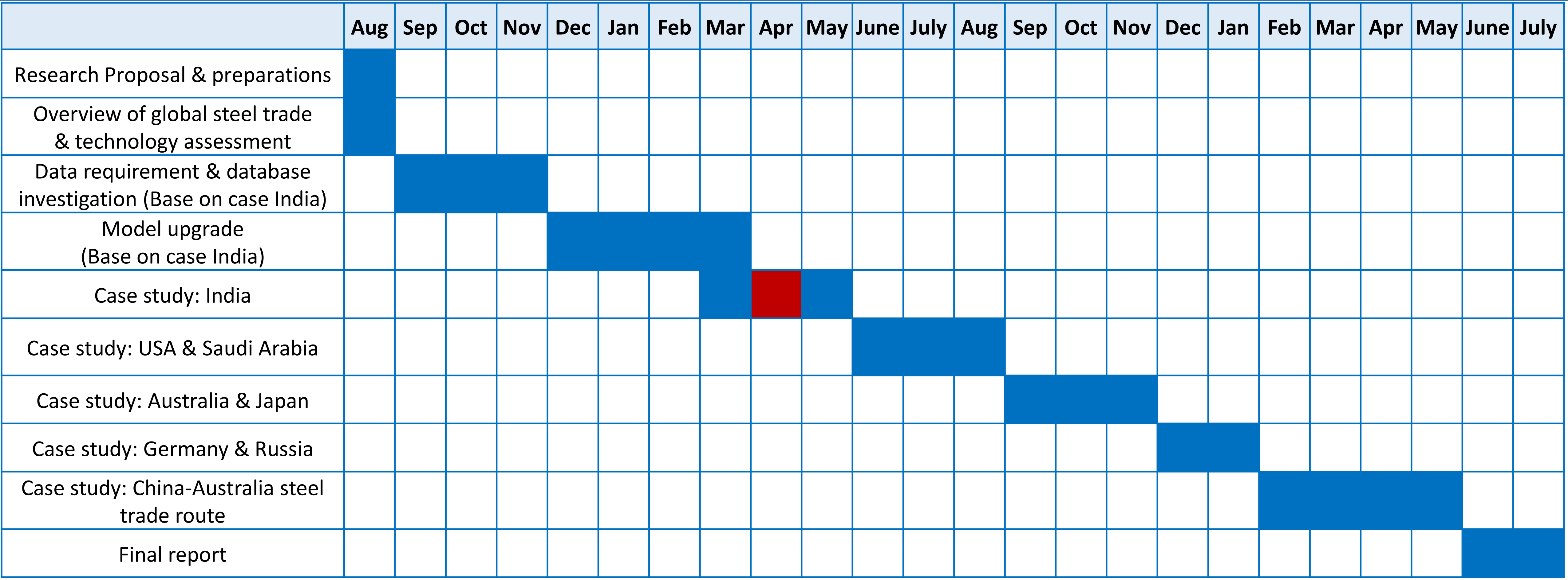
清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Presenter: Yunchuan Liu, PhD student

April 8, 2026

Project Timeline

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University



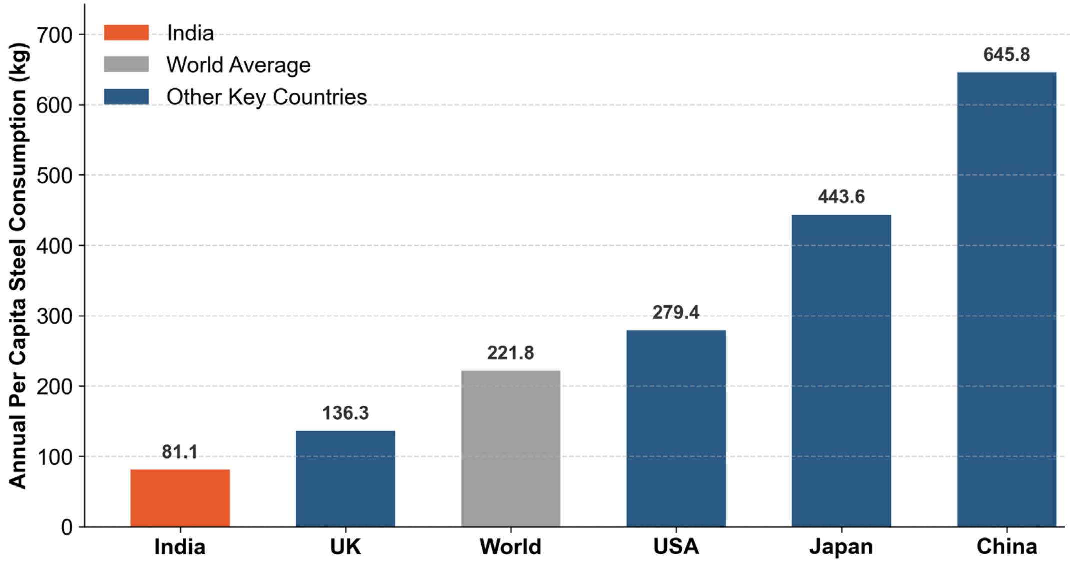
Research Update

- Review: Overview & Methodology: India's long-term iron & steel decarbonization planning
- India's Results (Demo)

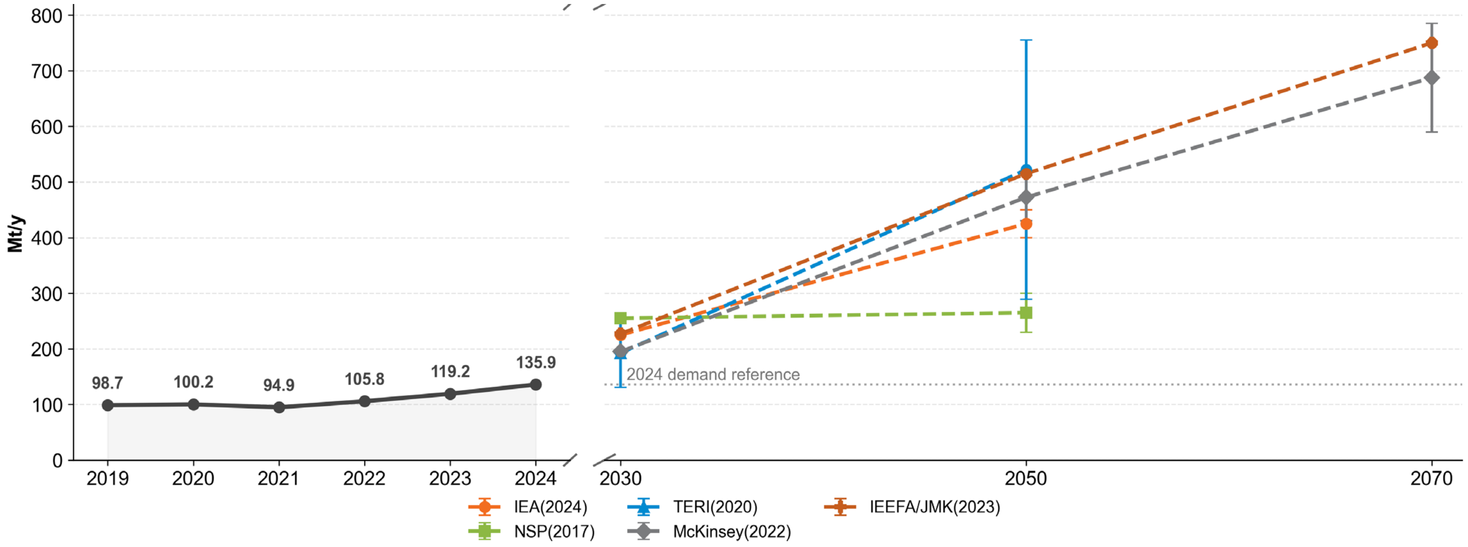
Case India: Review

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

India‘s I&S industry has huge development potential



Comparison of Annual Per Capita Steel Consumption: India vs. Key Countries

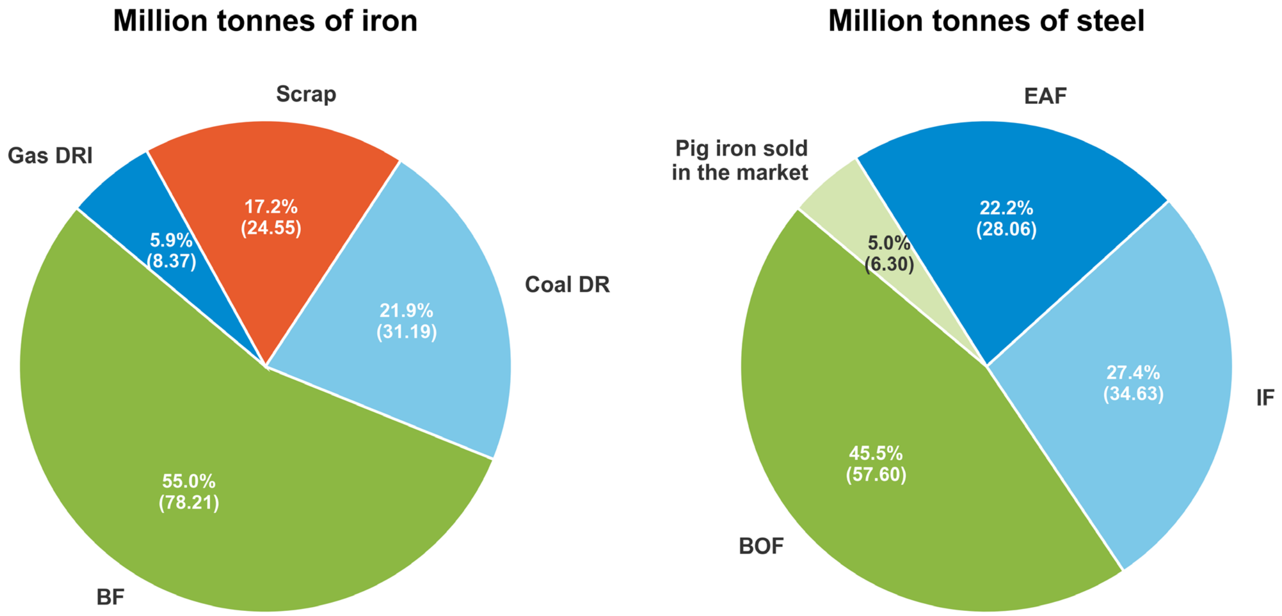


India's historical steel demand and institutional forecasts

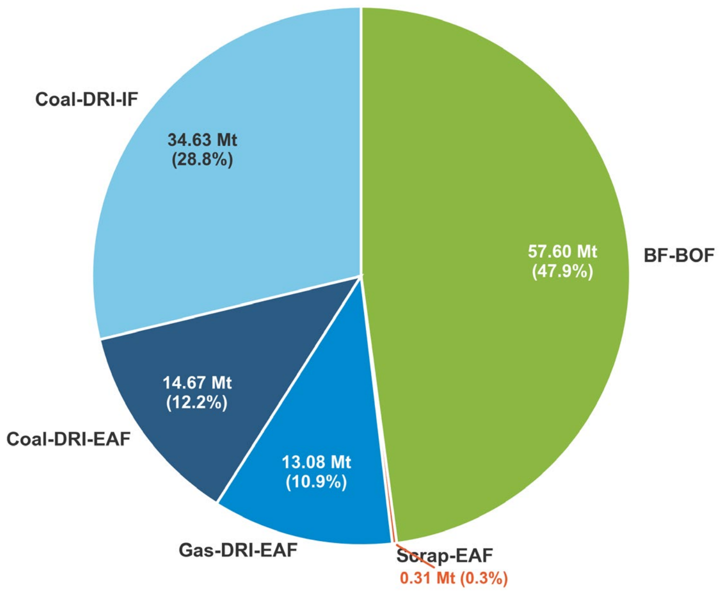
- India‘s per capita steel ownership is currently severely insufficient (**81.1kg vs 221.8kg world**), indicating a huge potential for growth in the steel market.
- Many institutional forecasts predict that India's steel demand will rise rapidly, reaching around 600~800 million tons by 2070.

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Main technical routes of the Indian I&S industry



India's iron and steel production technology structure (Mt)

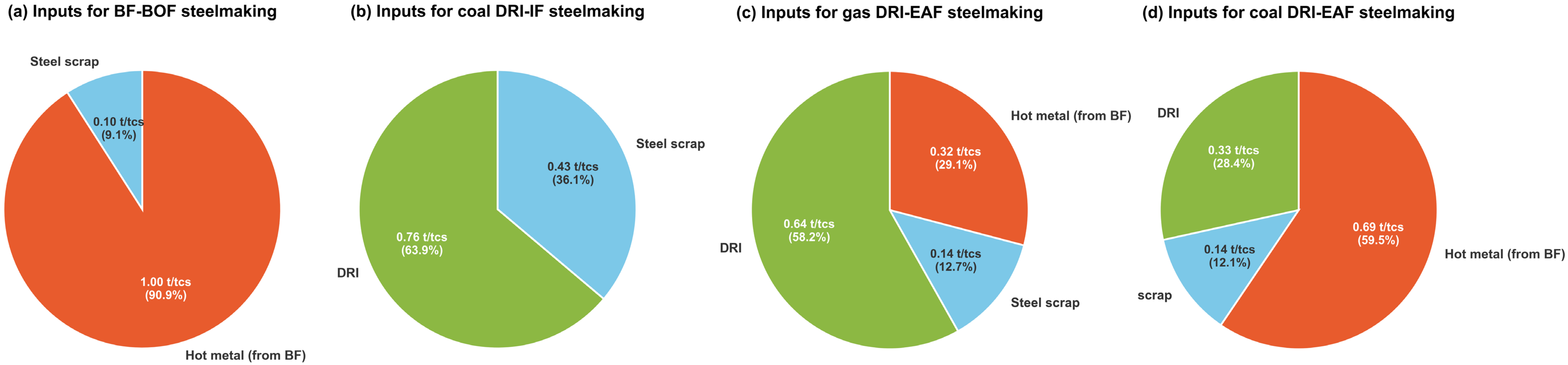


India's main iron & steel technical routes (Mt crude steel)

- India's ironmaking technology is primarily based on **BF(55%)** and **Coal-DRI(21.9%)**.
- Insufficient scrap steel supplement makes Scrap-EAF a minor tech route(0.3%) in India.

Source: BP, 2022

Main technical routes of the India I&S industry

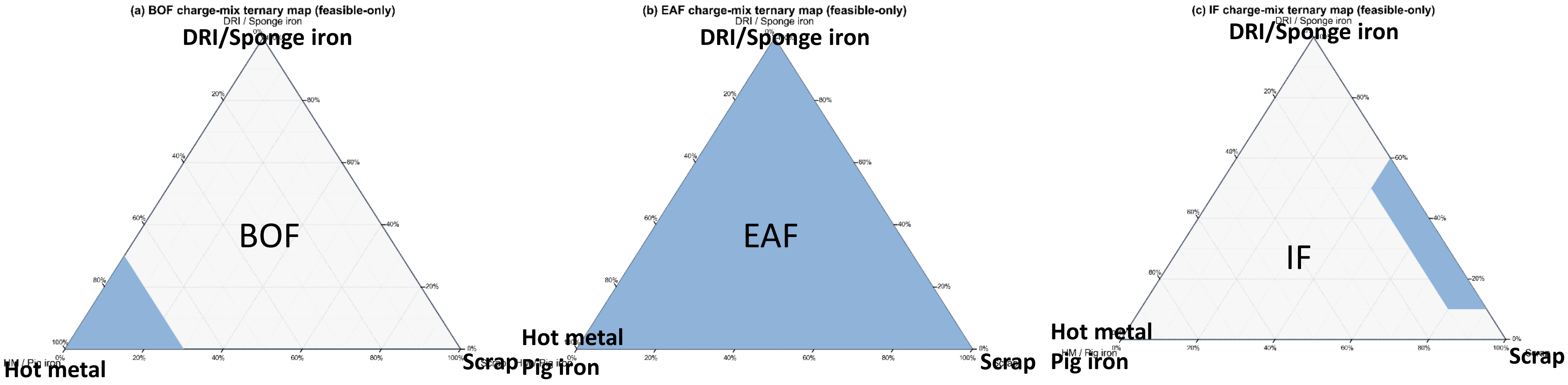


Typical input structures of various steelmaking technologies

- Hot metal from blast furnaces is widely used in BOF(91%), also can be mixed(30-60%) with DRI and Scrap in EAF.
- IF furnaces accept DRI and scrap, but cannot handle too much hot metal.

Source: BP, 2022

Steelmaking technologies

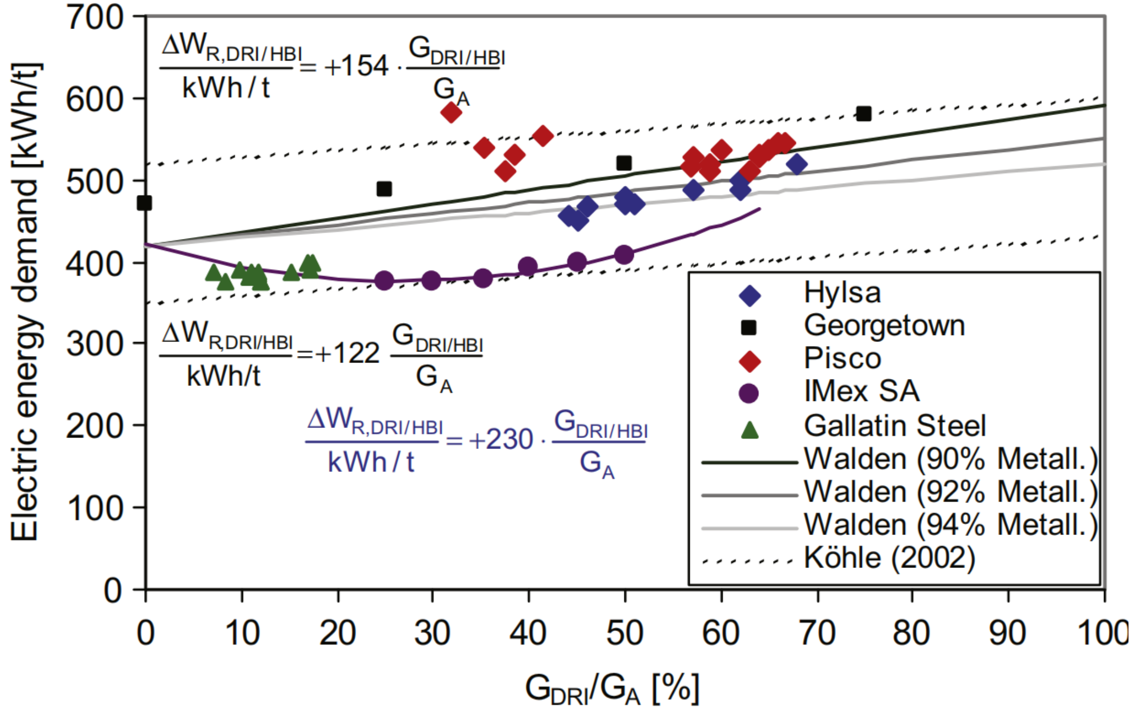


Available input structures of all steelmaking technologies in India

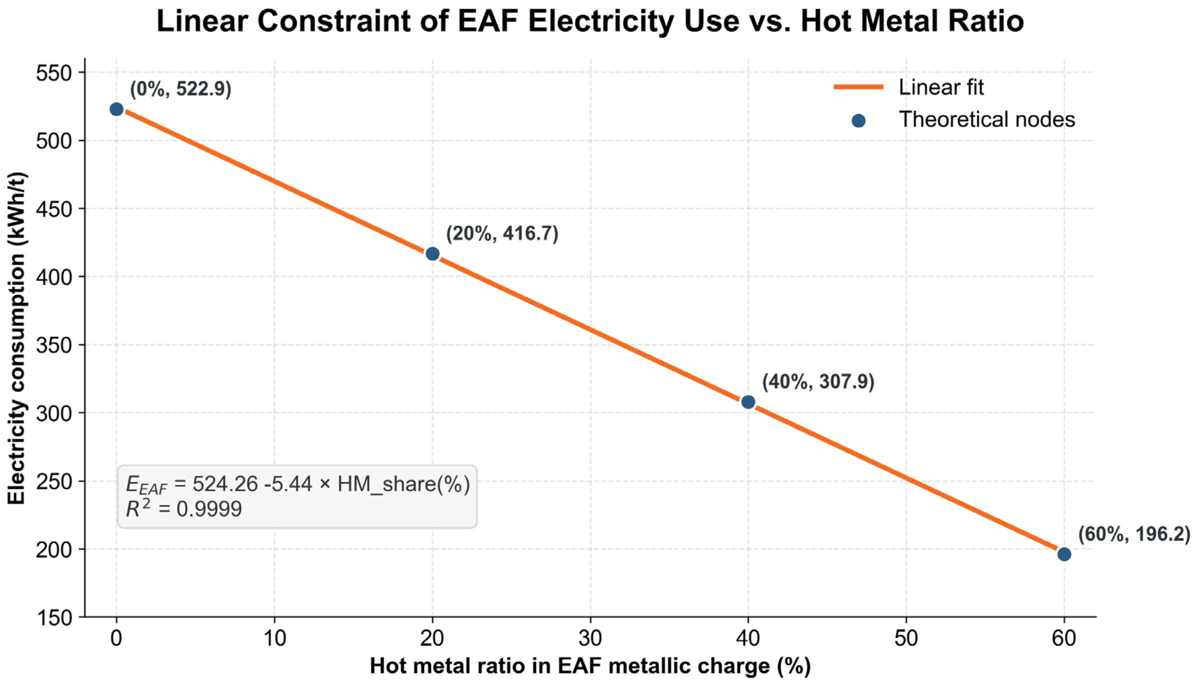
- In reality, production schedules are flexible, and a single furnace can **accommodate a range of input structures**.
- All furnaces are flexible, so enforcing a fixed input ratio may not make sense.
- As a result, the steelmaking and ironmaking processes should be decoupled.

Methodology

Steelmaking technologies



Relationship between **DRI ratio** and energy consumption in EAF furnace under various scenarios



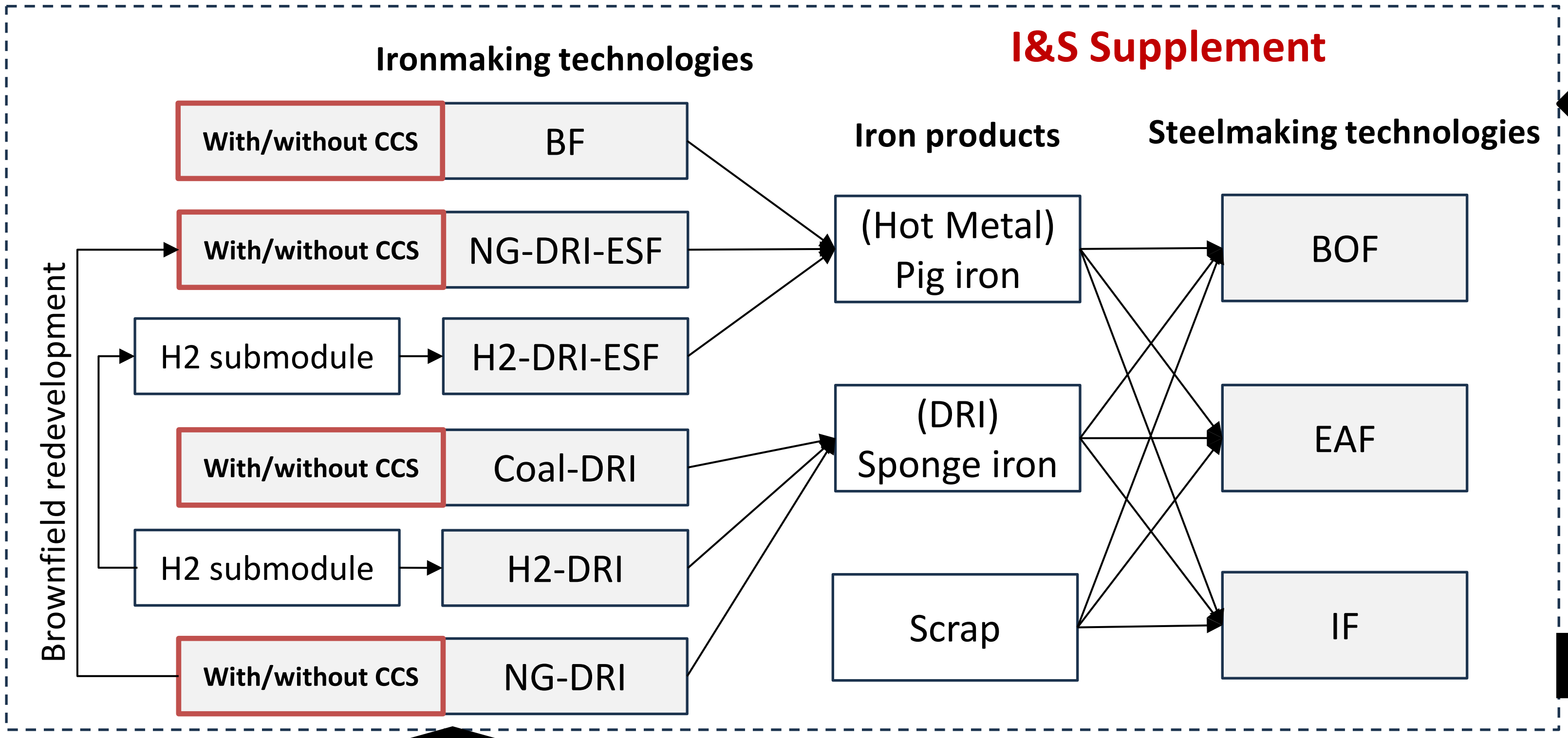
Relationship between **scrap ratio** and energy consumption in BOF furnace

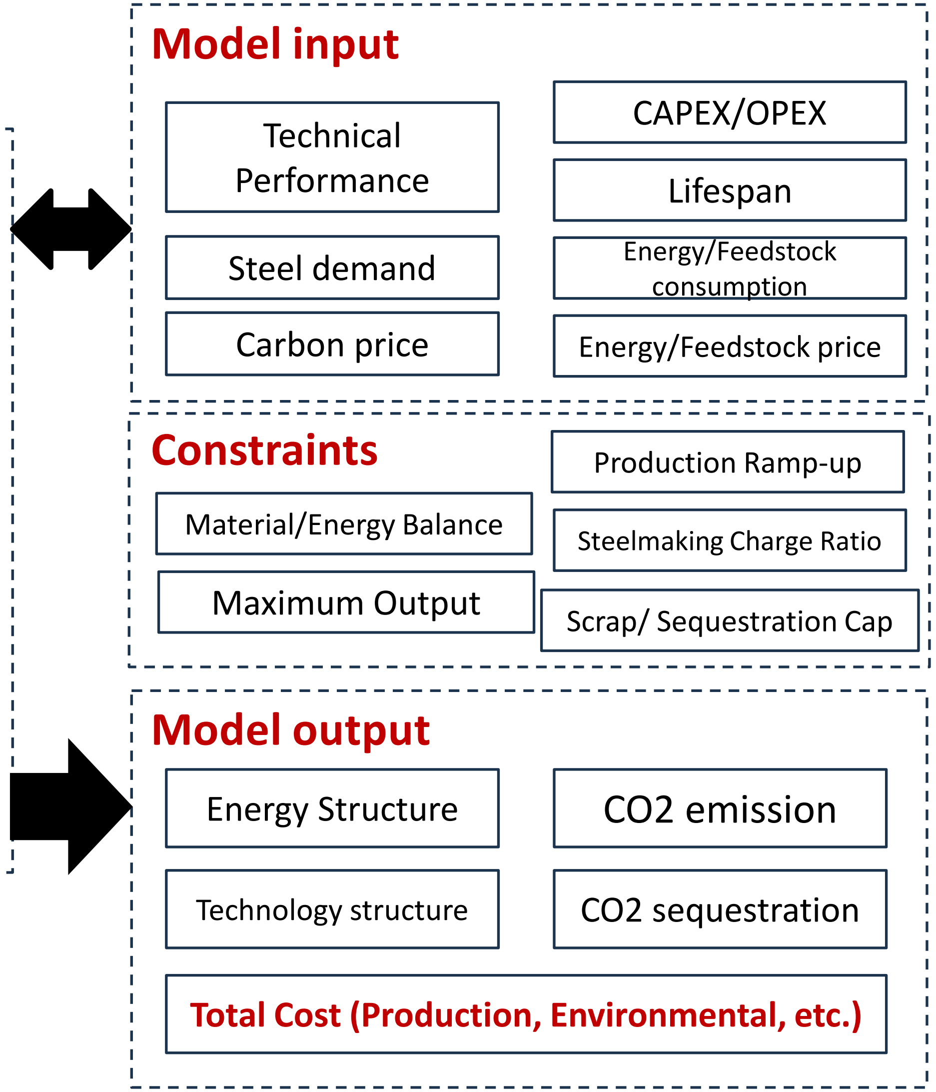
- The ratio of Scrap, Sponge iron, and Hot metal has an approximately linear effect on energy consumption.
- Formula used: For different types of steelmaking furnaces, there exists a set of α, β, θ such that –
  Energy consumption = α * hot_metal(%) + β * Sponge_iron(%) + θ * Scrap(%)

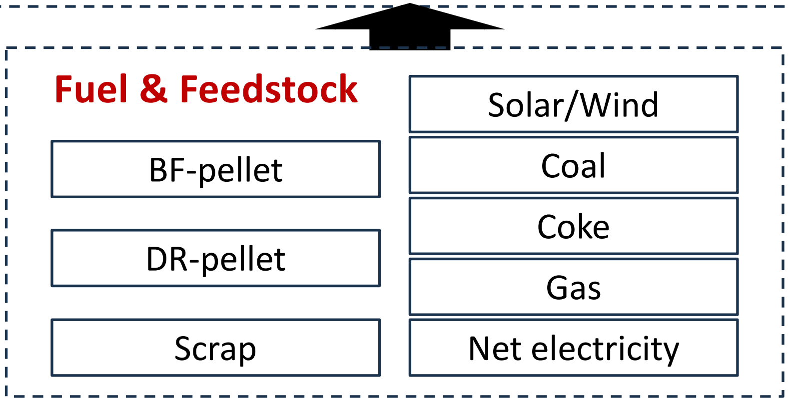
Source:
Kirschen et al. Influence of direct reduced iron on the energy balance of the electric arc furnace in steel industry, Energy, https://doi.org/10.1016/j.energy.2011.07.050
Hu et al. Carbon-material-energy flows nexus analysis of electric arc furnace: steel making processes in China, Journal of Cleaner Production, https://doi.org/10.1016/j.jclepro.2024.144377

Methodology

National planning Model





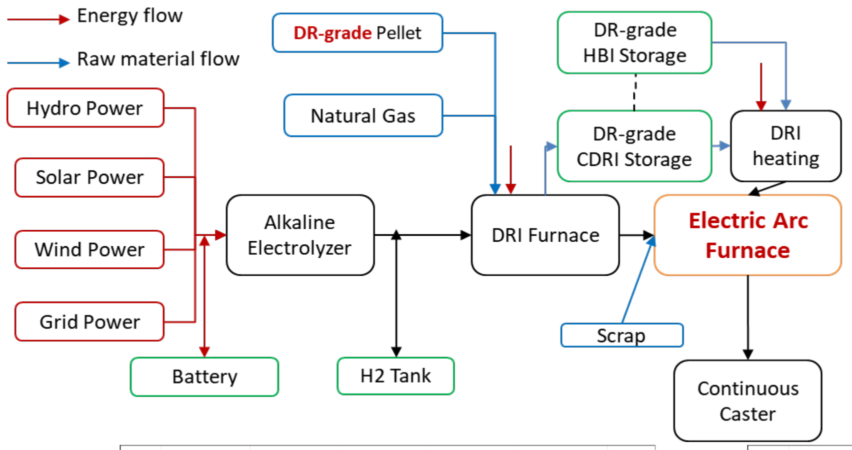


- Methodology: superstructure;
- Planning period: 2025~2070

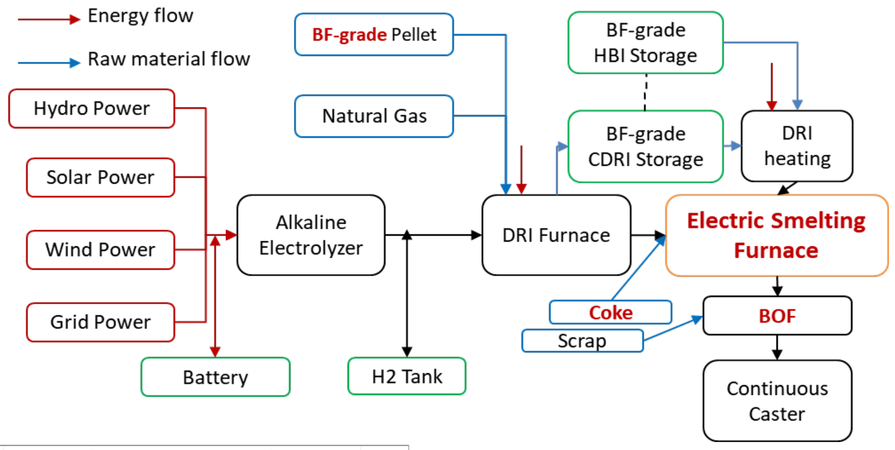
Link to the previous H2-DRI models

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Traditional H2-DRI-EAF route

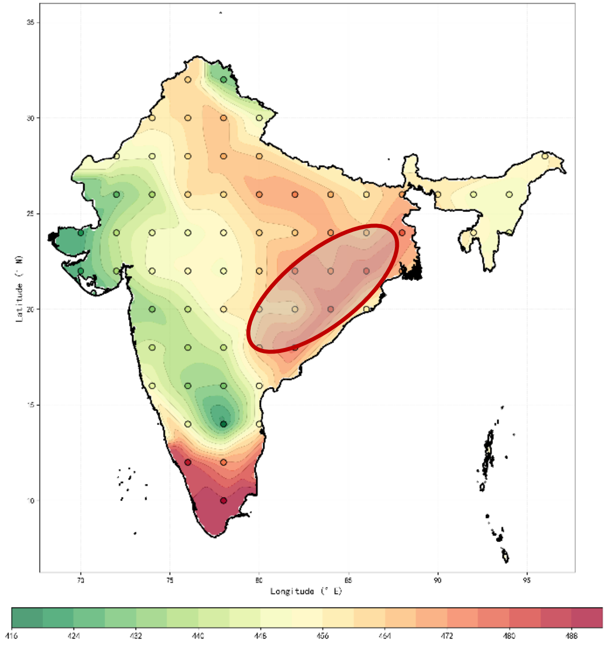


H2-DRI-ESF-BOF route

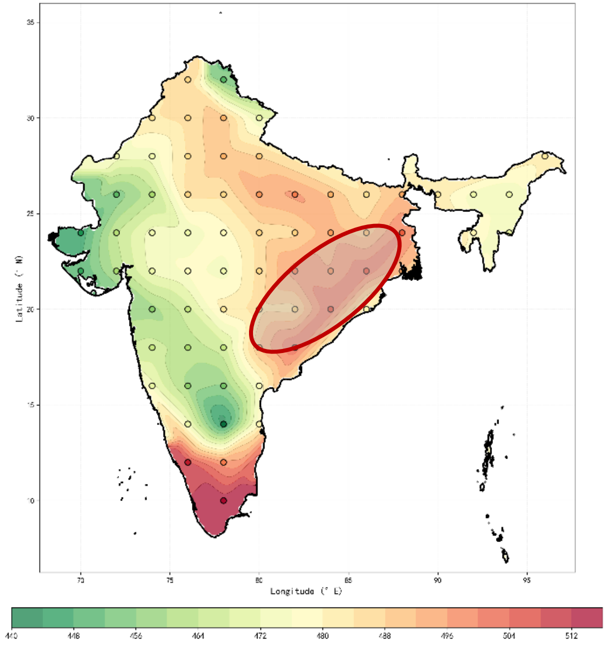


India's iron ore supply prices

Type	price	Source
import DR pellet	$158-168	Estimates based on the Platts pellet premium
import BF pellet	$112-118	CFR Kandla / Mundra
domestic BF pellet	$88-98	Odisha Index



H2-DRI-ESF Levelized cost of iron 416-488 USD/t



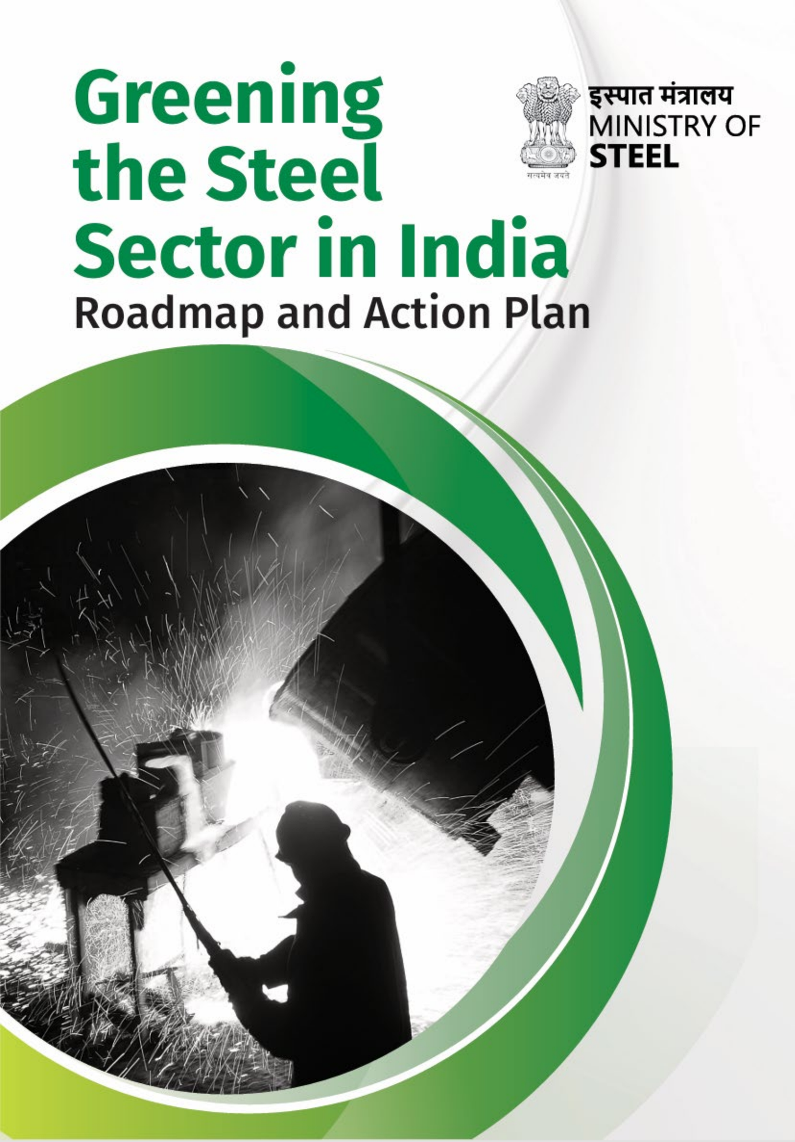
H2-DRI Levelized cost of iron 440-512USD/t

Model outputs

- Capacity
- Cost structure
- Production rhythm
- Levelized costs

- The results of the original model encompass the CAPEX for each technical stage of ironmaking, as well as the final LCOD.
- Soft Linkage: The cost structure of the original model is inherited by the ironmaking costs in the current model.

India’s Decarbonization policies



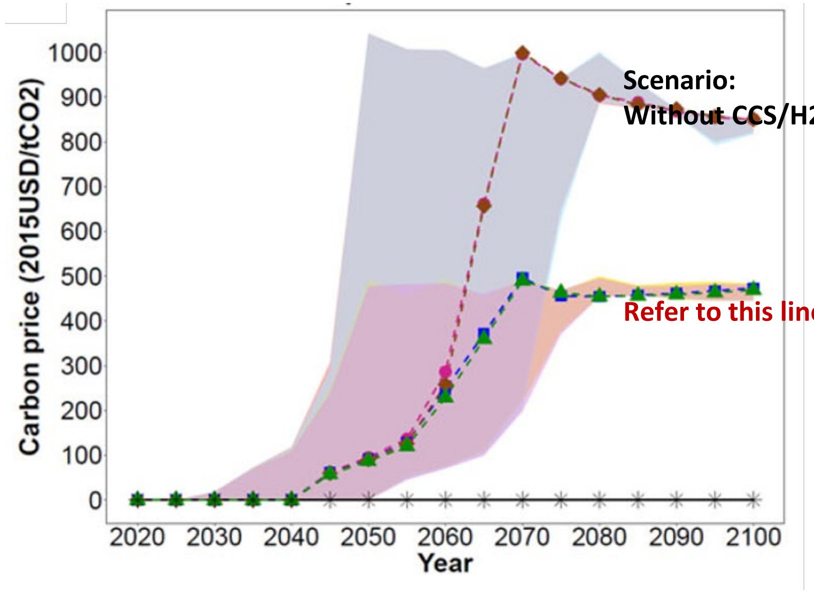
Timeline	Objective
2030	• Reduce integrated emission intensity to 2.20 tCO_2/tcs • 45% penetration rate of renewable energy • Significantly increase the utilization of scrap steel and pellet ore • Demonstrate pilot plants using green hydrogen
2070	• Achieve **net-zero emissions** for the steel industry

Q: What is the **transformation pathway** for India’s steel sector decarbonization plan (under different scenarios)?

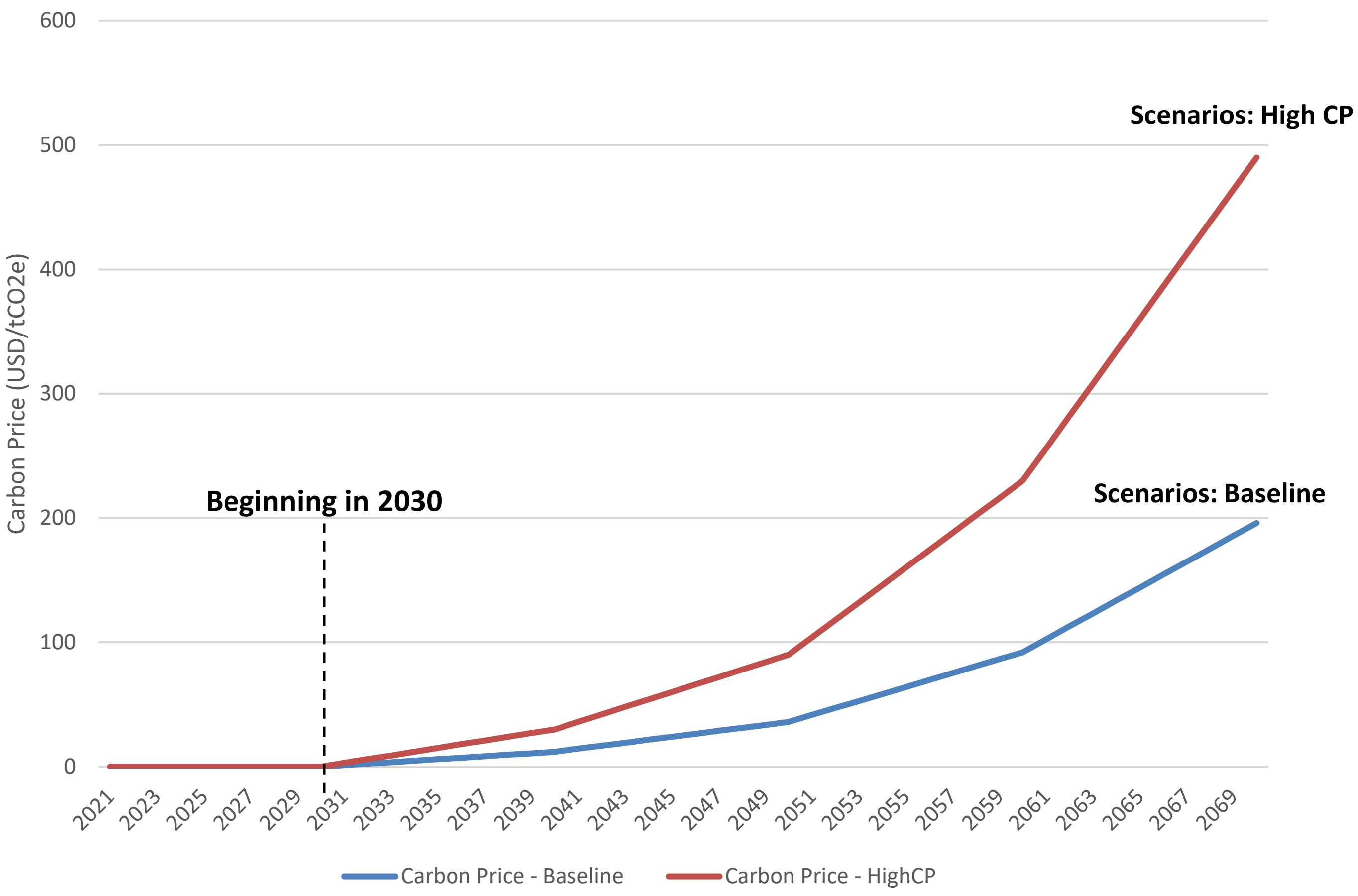
Case India: Scenario settings

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Important parameters: Carbon price



Ref. Carbon price across scenarios



Carbon Pricing under Different Scenarios (USD/tCO2e)

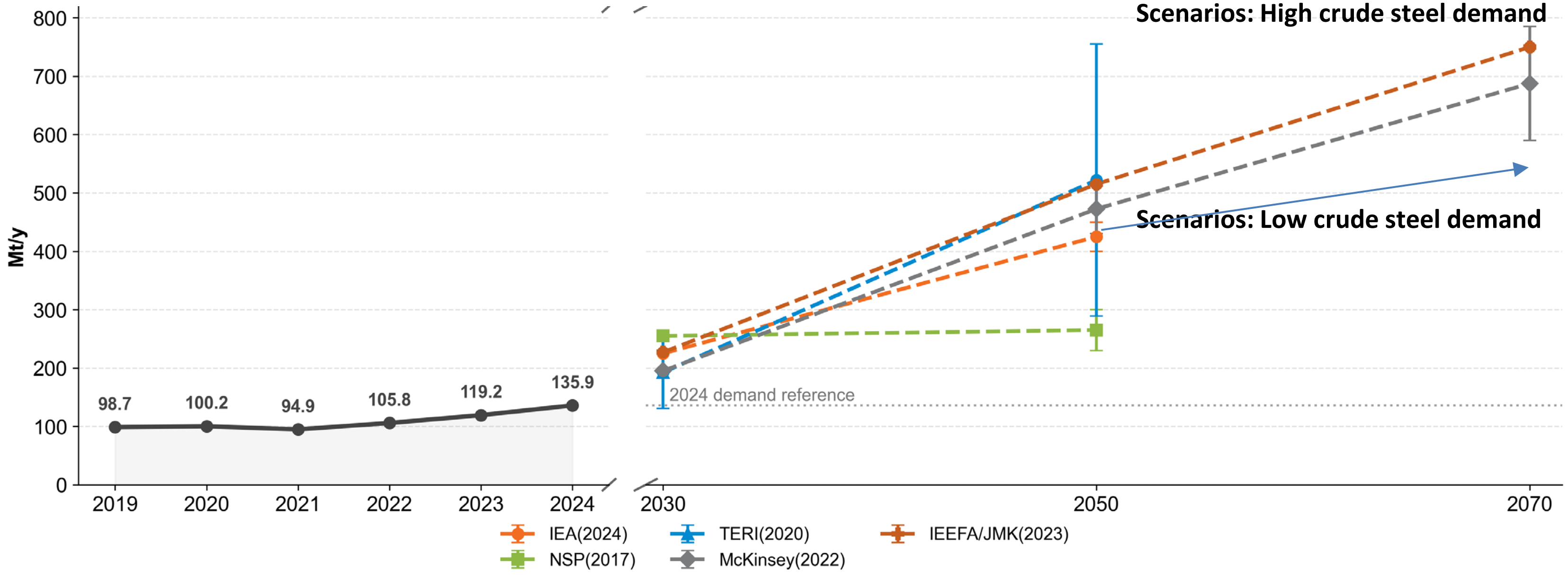
- Current analyses of carbon pricing in India are relatively limited.
- The approach currently refer to the research by CEEW, which used an integrated assessment model.
- We build a mild **baseline** with lower CP, and set **highCP scenarios** acting as the reference.

Source:
WEA India. Pricing carbon: Risks and opportunities for India – an analysis using India energy policy simulator
CEEW(Council on Energy, Environment and Water). Implications of a Net-Zero Target for India's Sectoral Energy Transitions and Climate Policy. doi.org/10.1093/oxfclm/kgac001

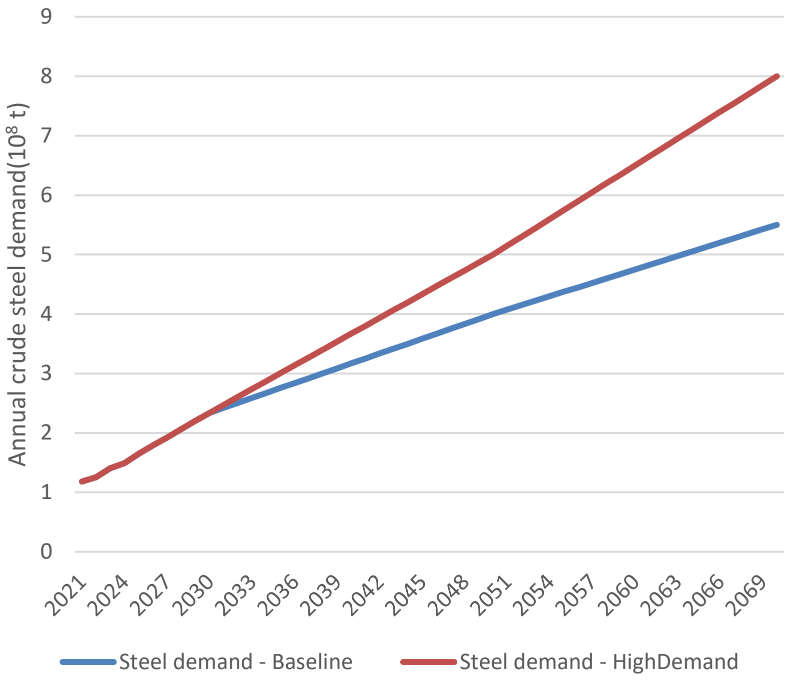
Case India: Scenario settings

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Important parameters: Steel demand



India's historical steel demand and institutional forecasts

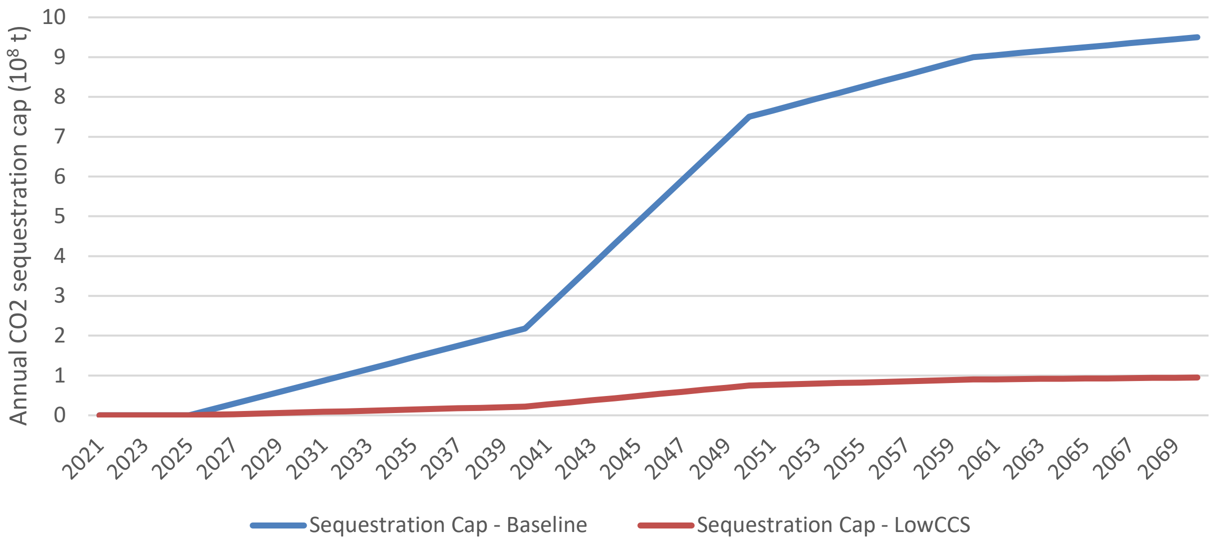


Demand settings under Different Scenarios

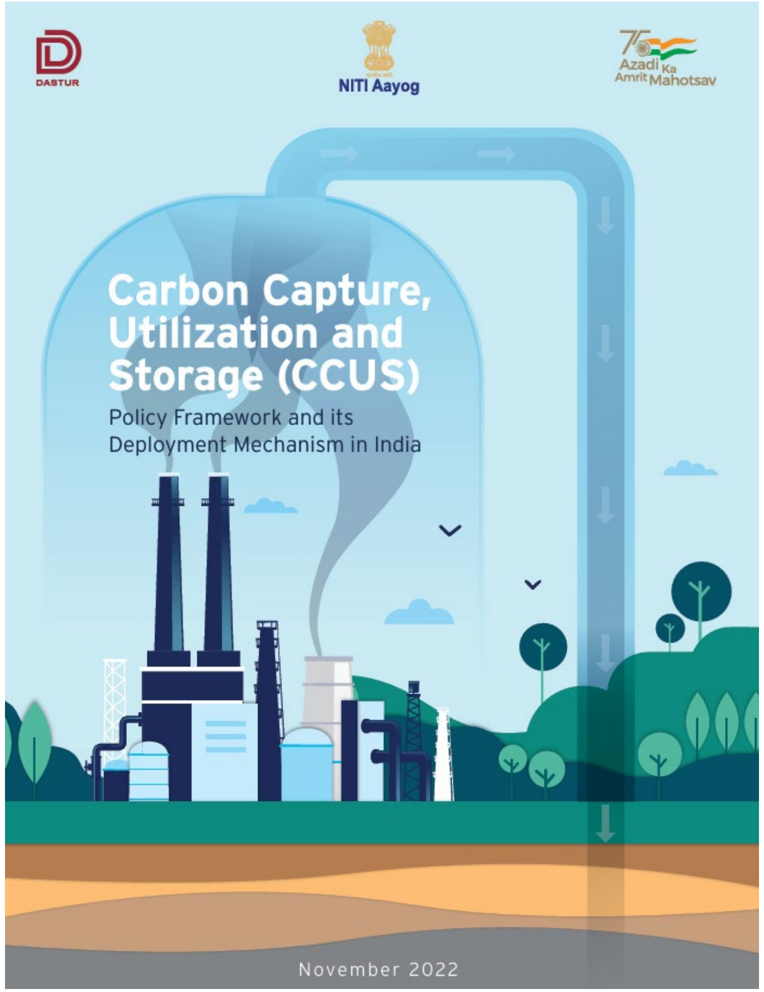
- The majority of institutional forecasts do not reach 2070.
- Forecasts looking ahead to 2070 suggests India's steel demand is expected to reach a level of **600 - 800 Mt**.
- We utilized the conservative IEA (2024) projection as our baseline, and the aggressive McKinsey projection as our high-demand scenario.

Case India: Scenario settings

Important parameters: CO2 sequestration limit



Parameter: CO2 sequestration cap



- Using the CEEW report which give the annual CO2 sequestration potential.
- **Baseline:** The reported cap of India's carbon sequestration capacity.
- **LowCCS:** The cap of carbon sequestration capacity after allocation based on the Indian iron&steel industry's share of carbon emissions (10%).

Source: CEEW. Carbon Capture, Utilization and Storage (CCUS): Policy Framework and its Deployment Mechanism in India

Case India: Scenario settings

Scenario settings

Parameter	Settings	
Carbon price	Baseline	HighCP
Steel demand	Baseline	HighDemand
CO2 sequestration limit	Baseline	LowCCS
Scrap / DR pellet / …	…	

- Combining the current parameter settings, we get a wide range of scenarios.
- For illustrative purposes, four specific scenarios (**BS**, **S1**, **S2**, and **S3**) have been selected here for comparative analysis.
- In the future, a new series of scenarios will be established by taking into account the prices and supply of scrap and DR pellets.

Scenario settings

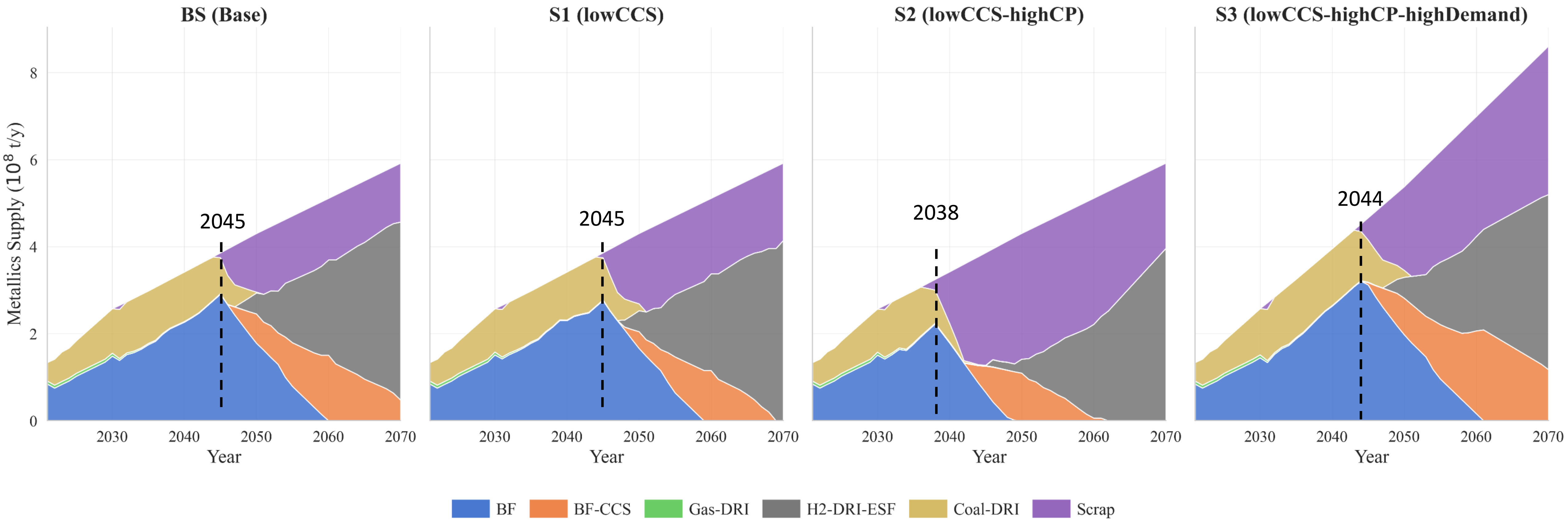
Base (BS)
lowCCS (S1)
highCP
highDemand
lowCCS-highCP (S2)
lowCCS-highDemand
highCP-highDemand
lowCCS-highCP-highDemand (S3)

BS-S1:
Lower sequestration limit

S1-S2:
Higher carbon price/tax

S2-S3:
Higher domestic steel demand

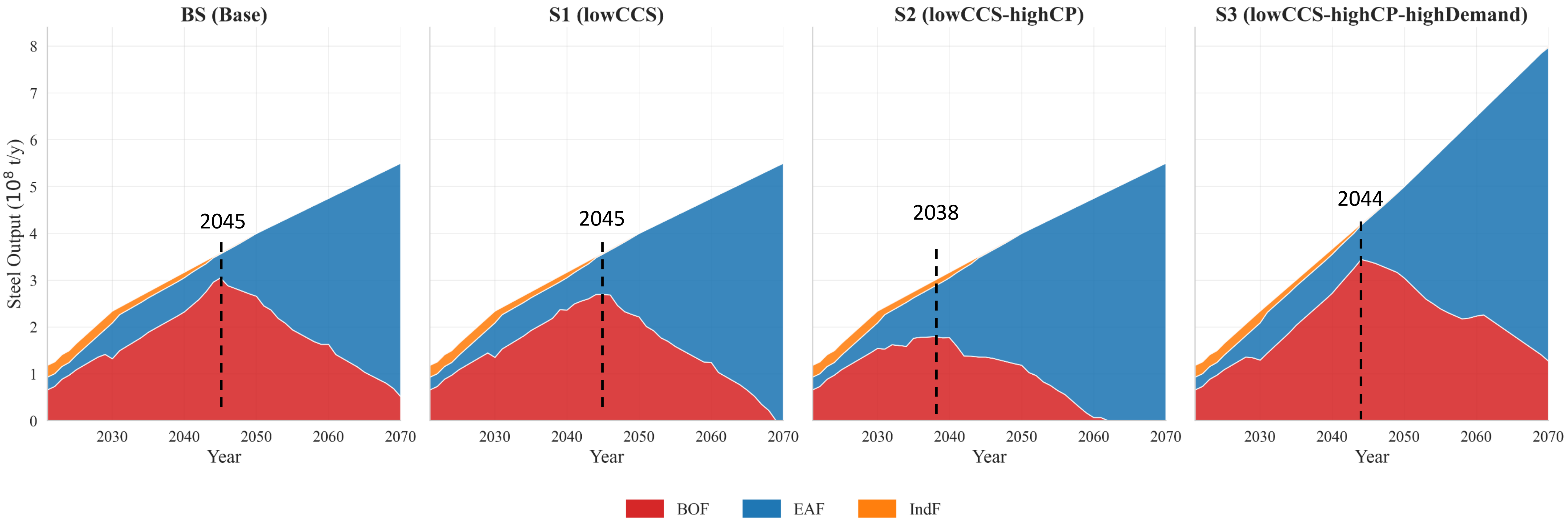
Technology structure: Ironmaking



- **Critical technological inflection points** (the production of Coal-DRI and BF start drpping): 2045 (BS, S1), 2038 (S2), 2044 (S3).
- Both low expectations for capacity expansion and high carbon prices serve to accelerate the restructuring of the technological structure.
- Ultimately, India's iron inputs for steelmaking will be primarily supplied by H2-DRI-ESF, scrap, and BF-CCS (sometimes).

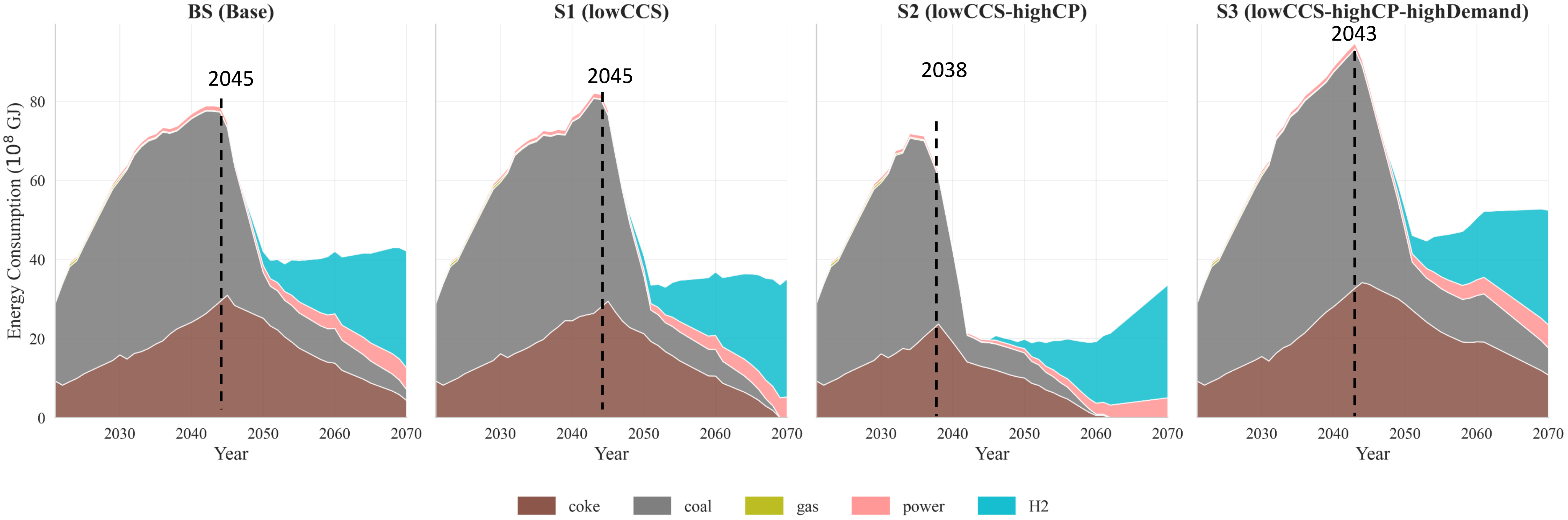
Results

Technology structure: Steelmaking



- Both expectations of limited capacity expansion and high carbon taxes can accelerate the restructuring of the technological landscape.
- Energy-intensive, low-yield IF capacity is expected to be naturally phased out around 2045 (mainly natural retirement).

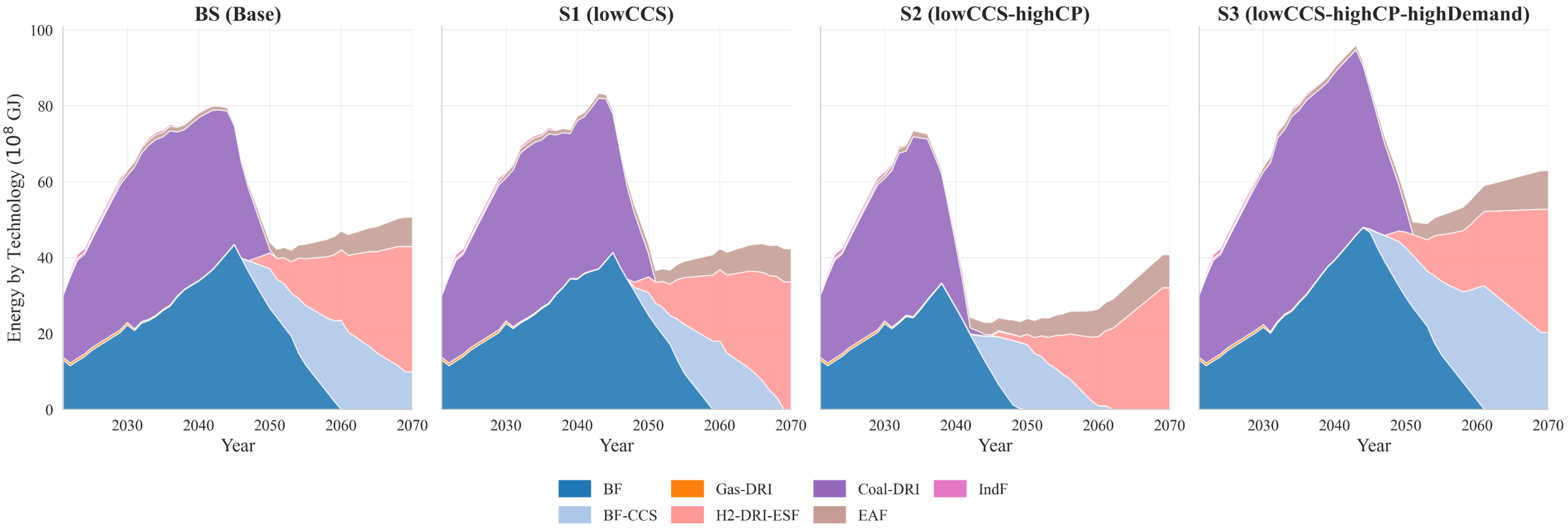
Energy consumption



- Before the technological inflection points, driven by new BF, energy consumption was dominated by coke and coal.
- After the technological inflection points, the consumption of coke and coal began to decline sharply, mainly replaced by hydrogen and power.

Results

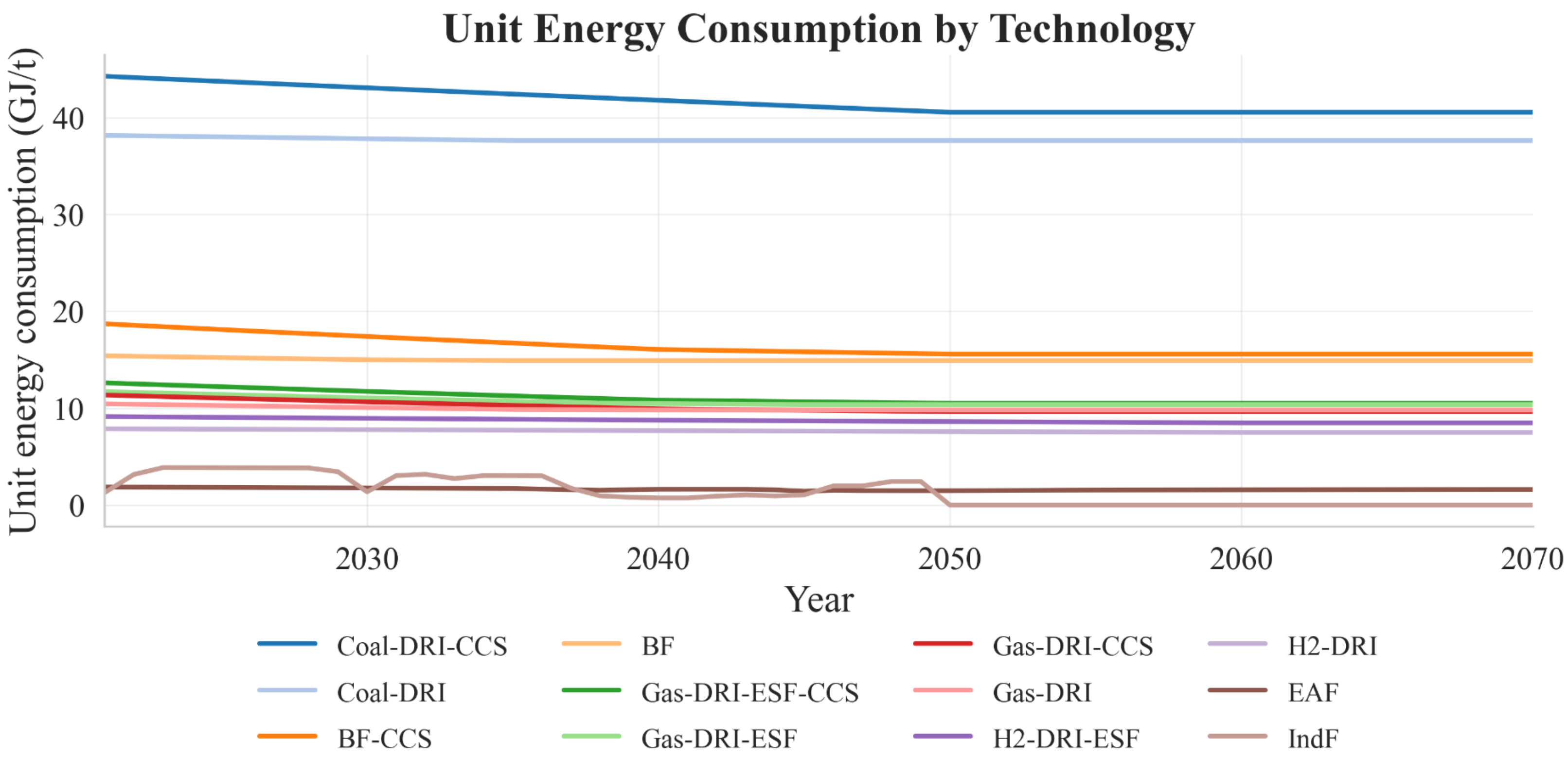
Energy consumption (by ironmaking and steelmaking technology)



- The great technology restructure: [BF & Coal-DRI] -> [EAF & H2-DRI-ESF]

Results

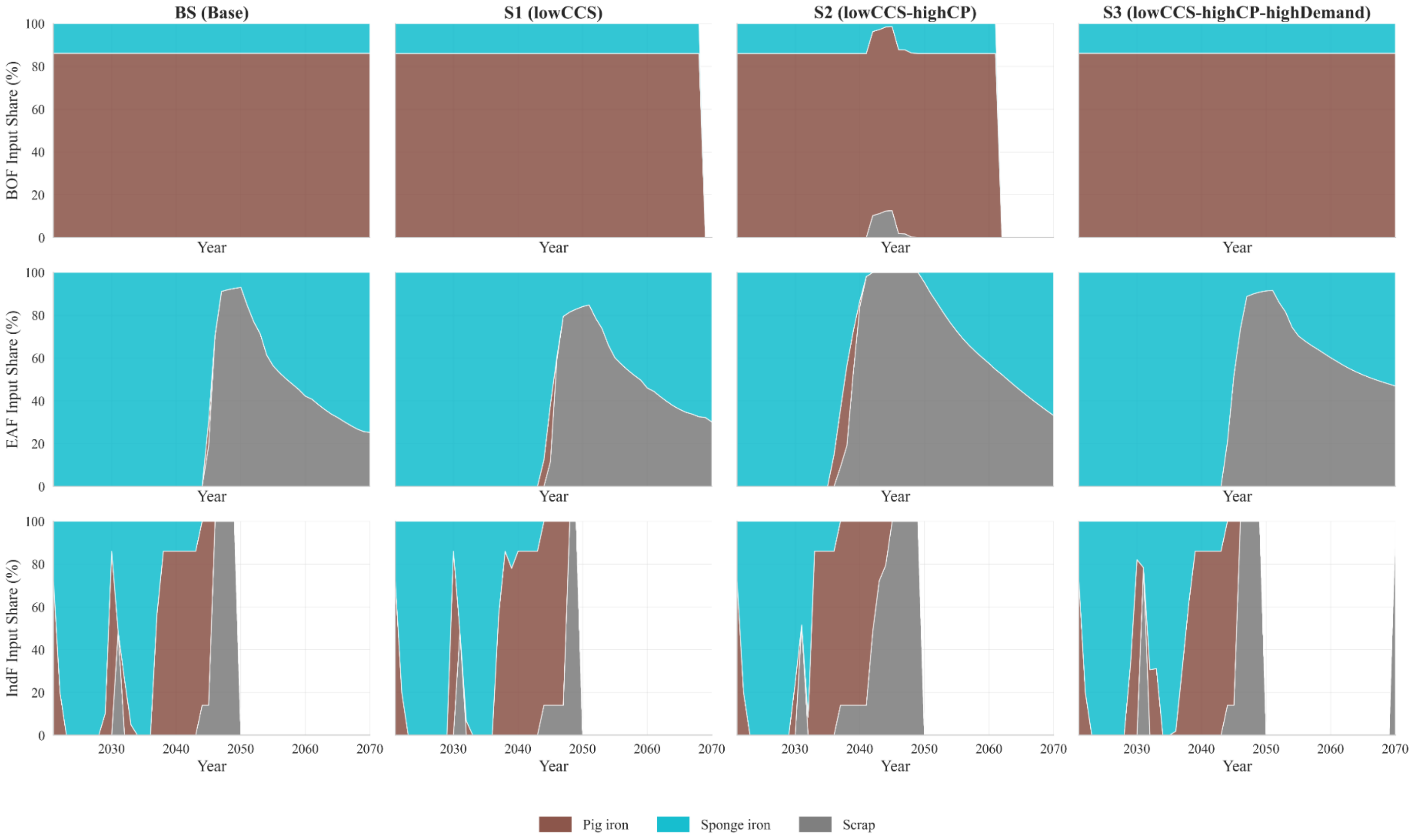
Unit energy consumption (by ironmaking and steelmaking technology)



- BF & Coal-DRI series are the most energy-ineffective ironmaking technologies.
- Steelmaking technologies' unit energy consumption depends on their iron input structure.

Results

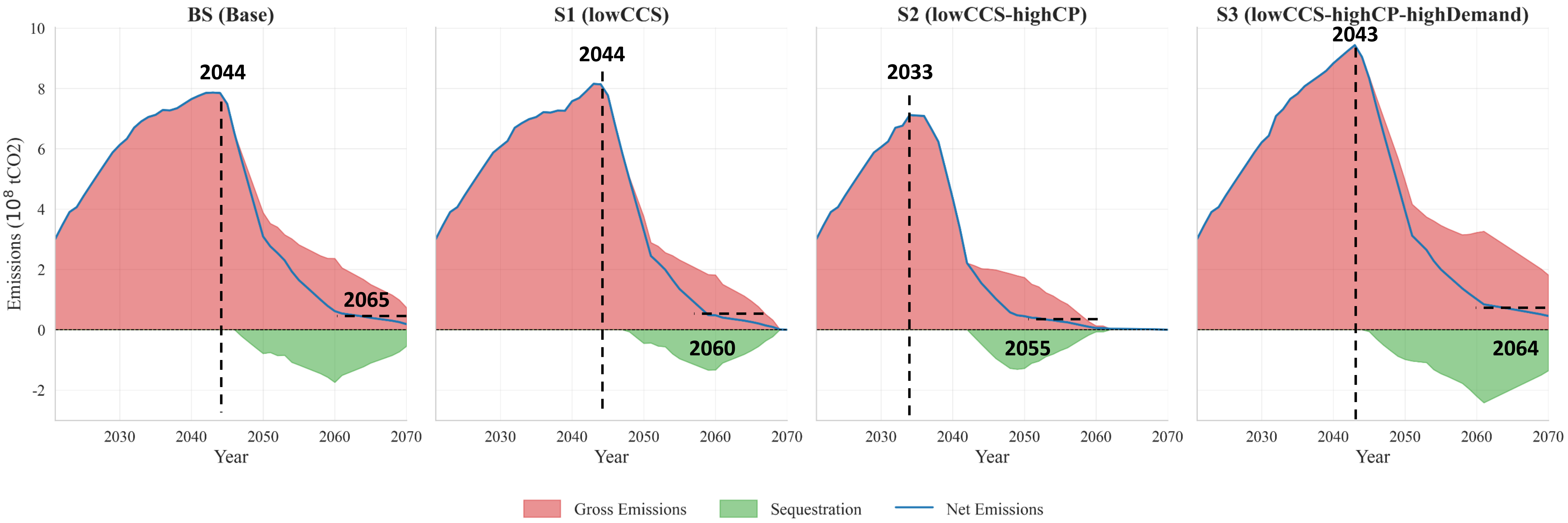
Iron input structure for steelmaking technologies



- The induction furnace seems to gain most dynamic input structures. (We assume IF is available for Pig iron but cost more energy)
- The blank area means the technology doesn't exist.

Results

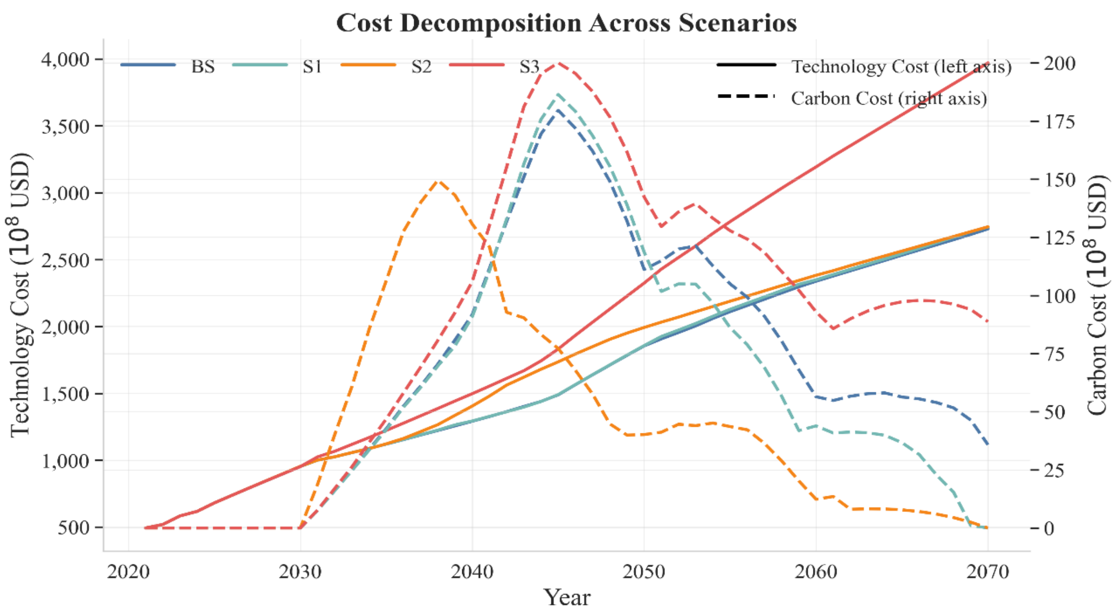
Emission & sequestration

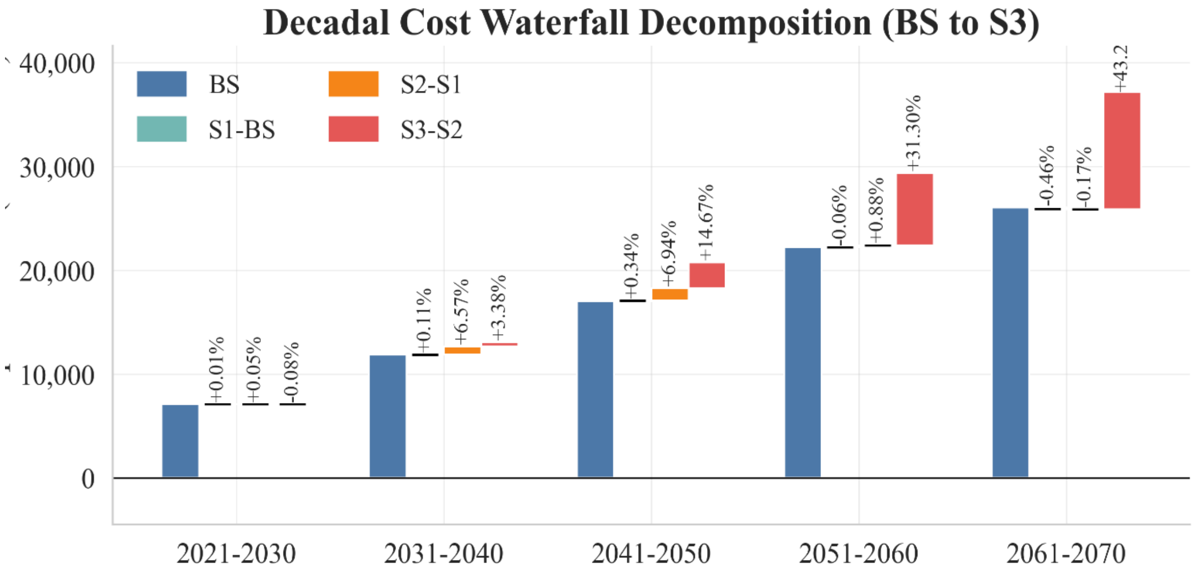


- Because of capacity expansion, the industry's total carbon emissions are projected to peak at about the technological inflection points.
- Under both scenarios, a 95% decarbonization (based on the peak, we view this as a manifestation of deep decarbonization) is achieved before 2070.
- Excessively high CCS quotas could actually slow down emissions reductions.

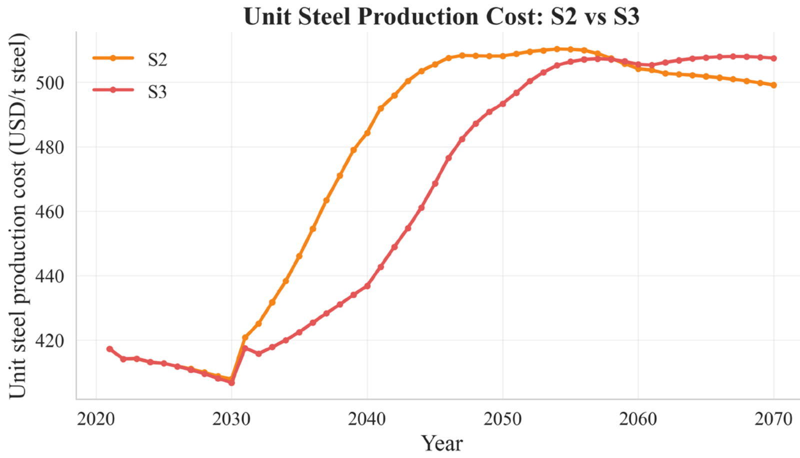
Results

Total cost





- Variations in the CO_2 sequestration limit within the normal range have a negligible impact on total costs.
- High carbon prices will raise total costs during the main technology transition period (2030–2050, >5%).
- The increased demand in S3 supported a higher proportion of new production capacity, reduce the unit steel production costs for a period.



Net steps

- More scenarios and further uncertainty analysis
  - Scrap price / limit
  - DR pellet price / limit
- Modeling and analysis of more countries

Thank You For Your Attention!